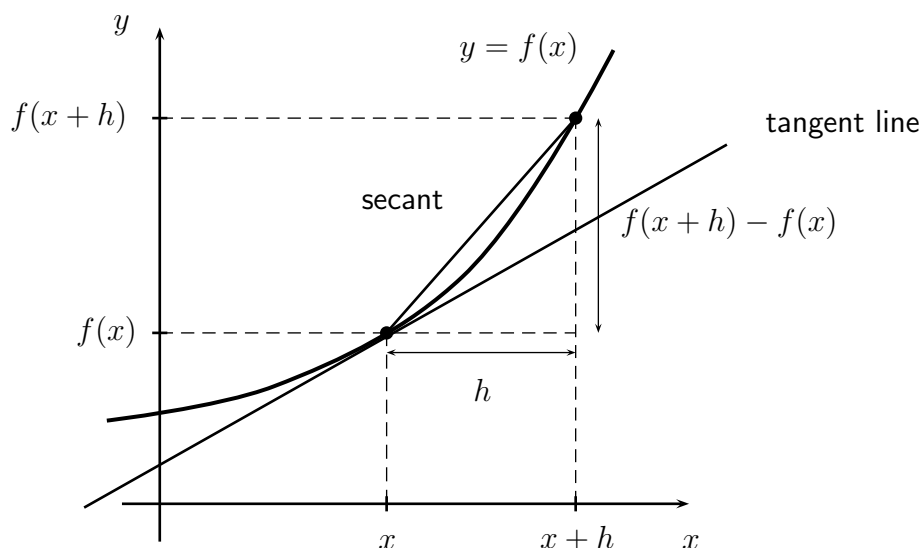


Notes for ‘Tangent lines and secants’

Important Ideas and Useful Facts:

- (i) **Smooth curves and tangent lines:** A curve $y = f(x)$ in the xy -plane is thought of as *smooth* if it can be approximated arbitrarily well at each point by a *tangent line*. To make this mathematically precise we need the notion of a *derivative* (explained later), which is defined in terms of *limits* (also explained later) of fractions. The fractions that arise are in fact slopes of secants, where a *secant* is a line segment joining two points on a curve.
- (ii) **Slopes of secants and tangent lines:** If a curve $y = f(x)$ is smooth then the slope of the tangent line at a given point of interest is approximated arbitrarily well by the slopes of lines (known as *secants*) formed by joining the point of interest to nearby points on the curve. By seeing what happens to the slope of the secant as the nearby point approaches the point of interest on the curve, we obtain the slope of the tangent line.



In the above diagram the slope of the secant is

$$\frac{f(x+h) - f(x)}{h}$$

and this approaches the slope of the tangent line as h approaches 0.

- (iii) **Slope of the tangent line to a parabola:** The slope of the tangent line to the parabola

$$y = kx^2$$

where k is a constant at the point (x, kx^2) is $2kx$, explained below.

In particular, the slopes of the tangent line to the parabolas $y = x^2$ and $y = -x^2$ at the points (x, x^2) and $(x, -x^2)$ are $2x$ and $-2x$ respectively.

To see why, consider $y = f(x) = kx^2$, where k is constant. Consider a small nonzero perturbation h . Then the slope of the secant joining the points

$$(x, f(x)) = (x, kx^2) \quad \text{and} \quad (x+h, f(x+h)) = (x+h, k(x+h)^2)$$

on the curve is

$$\begin{aligned}\frac{f(x+h) - f(x)}{h} &= \frac{k(x+h)^2 - kx^2}{h} = \frac{k(x^2 + 2xh + h^2 - x^2)}{h} \\ &= \frac{k(2xh + h^2)}{h} = \frac{hk(2x + h)}{h} = k(2x + h) .\end{aligned}$$

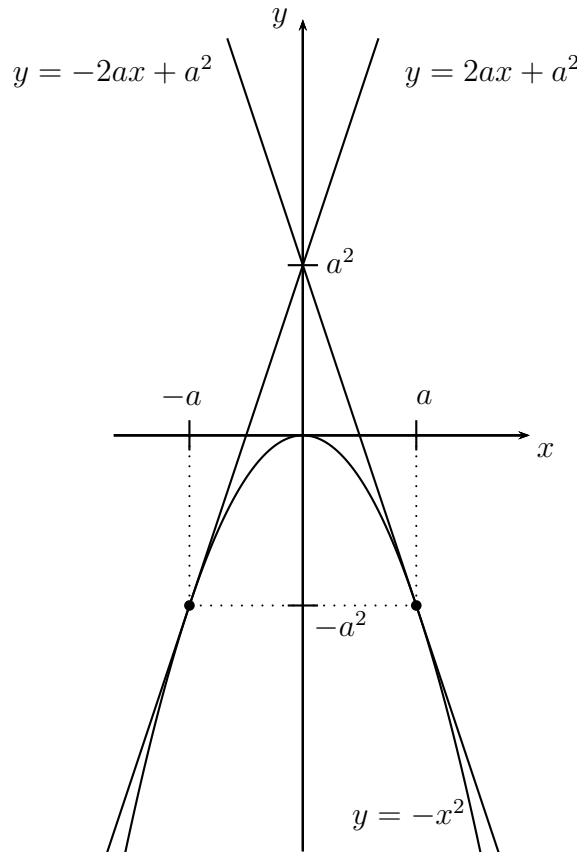
As h approaches 0, the slope of the secant approaches the slope of the tangent line to the curve at the point (x, kx^2) . As h approaches 0, this slope $k(2x + h)$ approaches

$$k(2x + 0) = k(2x) = 2kx .$$

This completes the verification that the slope of the tangent line is $2kx$.

Examples:

1. The slope of the tangent line to the parabola $y = -x^2$ at $x = 0$ is $-2 \times 0 = 0$, as expected, as the point $(0, 0)$ is the apex of the parabola and the x -axis is the tangent line at the apex, which is horizontal of slope 0.



The slopes of the tangent lines to the parabola $y = -x^2$ at $x = a$ and $x = -a$ are $-2 \times a = -2a$ and $-2 \times (-a) = 2a$ respectively. The tangent lines to the parabola at the points $(a, -a^2)$ and $(-a, -a^2)$ have equations $y = -2ax + a^2$ and $y = 2ax + a^2$ respectively (as one can quickly check) and they always intersect at the common y -intercept of a^2 .

2. The slopes of the tangent lines to the parabola $y = \pi x^2$ at $x = 2$ and $x = 10$ are 4π and 20π respectively, which are the perimeters of circles of radius 2 and 10 respectively.